MIL-E-1/1577B
25 SEPT 1975
SUPERSEDING
MIL-E-1/1577A
14 February 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 5973

This specification is approved for use by all Departments and Agencies of the Department or Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Clipper diode, rectifier, half wave

See figure 1

Mounting position: Vertical, cathode and base down

Weight: 3-pounds nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	Ef V	epx kv	ib a	Io Adc	Ieff Aac	tk sec	Pp W	TE °C	Base temperature °C
Rectifier (1):	See note 1		See note 2		See note 3			See note 4	See note 4
Maximum:	16.8	75	3.14	1.00	---	---	800	300	150
Minimum:	15.2	---	----	----	---	30	---	---	---
Rectifier (2):									
Maximum:	16.8	40	3.92	1.25	---	---	850	300	150
Minimum:	15.2	---	----	----	---	30	---	----	---
Clipper diode:									
Maximum:	16.8	33	20.0	----	2.0	---	850	300	150
Minimum:	15.2	--	----	----	---	30	---	---	---
TEST CONDITIONS:	16.0	--	----	----	---	30	---	---	---

GENERAL:

Qualification - Required

5973

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						Min	Max	
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	18.0	20.2	A
1261	Anode voltage	ib = 5.0 a See note 6	0.65	II	eb	850	1,050	v

(B) 1353	Operation of rectifiers (1)	epx = 75 kv;
		ib = 3.14 a;
		Io = 1.0 Adc;
		t = 600 seconds
		See note 7
			0.65	II	---	---	---	---

(B) 1353	Operation of rectifiers (2)	epx = 40 kv;
		ib = 3.92 a;
		Io = 1.25 Adc;
		t = 600 seconds
			0.65	II	---	---	---	---

1231	Peak emission	eb = 4 kv (max) See note 6	0.65	II	ib	20	---	a
	Quality conformance inspection, part 3							
----	Service life	t = 1,000 hours	---	---	---	---	---	---
(B) 1353	Operation of rectifiers (3)	Ieff = 2.2 Aac; t = 600 seconds See notes 5 and 8	---	---	---	---	---	---

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NOTES:

1. Filament voltage must be applied so that filament current does not exceed 30 amperes, even momentarily.

2. Values of peak anode current up to 5 amperes may be used in rectifier service provided anode dissipation ratings are not exceeded. For 800 watts dissipation, the maximum anode temperature should not exceed 985°C as observed with an optical pyrometer. For 850 watts, this temperature should not exceed 1010°C.

3. True rms value of the sum of the fundamental and all harmonic components of anode current.

4. Where tubes are operated in close proximity or in enclosed equipment, sufficient forced air must be provided to keep bulb and bases below rated maximum temperature.

5. The tube shall be operated as a clipper diode in a line type thyratron modulator under the following conditions: epx between 33 kv and 40 kv (adjust to give Ieff = 2.2 Aac); ib = 20 a minimum; network impedance Zo = 8.3 ohms; RL = 5.5 ohms; tp = 2.5 μs; prr = 360; clipper diode series resistance = 125 ohms. The tube shall operate without evidence of arcing for the specified time.

6. Voltage must be applied in pulses so that tube will not be damaged.

7. Tube shall operate without evidence of arcing for specified time following a preliminary operating period not to exceed 10 minutes.

(B) 8. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of two. In the event of failure the test will be made as a part of quality conformance inspection (QCI), part 3 code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Preparing activity:

Air Force - 85

Review activities:

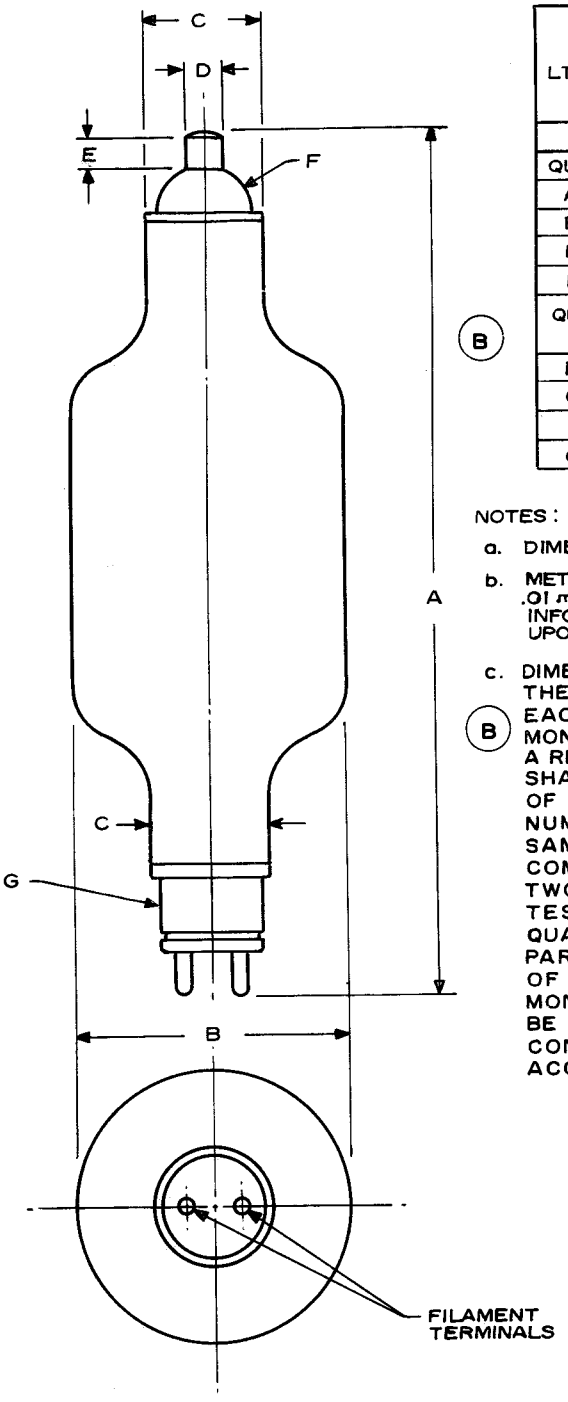
Air Force - 17, 80
DSA - ES

(Project No. 5960-2850)

User activities:

Army - MU
Navy - WP, MC, CG
Air Force - 11, 14

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LTR	Dimensions in inches with metric equivalents (mm) in parentheses (see note b)	
	Minimum	Maximum
QUALITY CONFORMANCE INSPECT., PART 1		
A	18.74 (476.00)	19.50 (495.30)
B		6.125 (155.58)
D	.793 (20.14)	.807 (20.50)
E	.713 (18.11)	
QUALITY CONFORMANCE INSPECT., PART 3 SEE NOTE C		
F	CAP: C1-35 OR C1-10	
G	BASE: A2-87	
REFERENCE DIMENSIONS		
C	2.625 (66.68)	

NOTES:

a. DIMENSIONS ARE IN INCHES.

b. METRIC EQUIVALENTS (TO THE NEAREST .01 mm) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 mm.

c. DIMENSIONS SHALL BE CHECKED DURING THE INITIAL PRODUCTION AND ONCE (B) EACH SUCCEEDING 12 CALENDAR MONTHS IN WHICH THERE IS PRODUCTION. A REGULAR DOUBLE SAMPLING PLAN SHALL BE USED, WITH THE FIRST SAMPLE OF THREE TUBES WITH AN ACCEPTANCE NUMBER OF ZERO, AND A SECOND SAMPLE OF THREE TUBES WITH A COMBINED ACCEPTANCE NUMBER OF TWO. IN THE EVENT OF FAILURE, THE TEST WILL BE MADE AS A PART OF QUALITY CONFORMANCE INSPECTION, PART 2, CODE LEVEL D, WITH AN AQL OF 6.5. THE REGULAR "12 CALENDAR MONTH" DOUBLE SAMPLING PLAN SHALL BE REINSTATED AFTER THREE CONSECUTIVE SAMPLES HAVE BEEN ACCEPTED.

FIGURE 1. Outline drawing of electron tube type 5973

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